Chinmoy Mitra

Location: Dhaka, Bangladesh | **Contact No:** +88 01708770617

Portfolio | rudrochinmoy@gmail.com | Google Scholar | GitHub | LinkedIn | LeetCode

Education

Rajshahi University of Engineering and Technology (RUET) May 2024
Bachelor of Science in Computer Science and Engineering *Rajshahi, Bangladesh*

- **Academic Performance:** Final Semester GPA: **3.70**/4.00 | Last 60 Credits: ∼**3.35**/4.00 | Overall CGPA: 3.11/4.00
- **Standardized Tests: GRE:** 313 (**Quant: 167**, Verbal: 146, AWA: 3.5) | **IELTS: 7.0** (L: 8.0, R: 7.0, W: 6.5, S: 7.0)
- **Relevant Coursework:** Artificial Intelligence, Neural Networks and Fuzzy Systems, Digital Image Processing, Linear Algebra with Computational Applications, Data Structures and Algorithms, Statistics and Probability

Publications

- **C. Mitra** and T. Mahmud. "**Beyond Aggregate Accuracy: Item-Level Transitions in Prompt Optimization**," *arXiv preprint*, 2026. [Submitted to ACL ARR 2026 (Oct Cycle)] arXiv
- M. M. H. Nipu, **C. Mitra**, *et al.* "**Trust-Gated Capability Control: Breaking the Trust-Vulnerability Paradox in Multi-Agent LLM Systems**" [Presented at IEEE iCONNECT 2026] arXiv
- **C. Mitra**, *et al.* "**Robust and Personalized Federated Learning for Aircraft-Engine Prognostics under Benign and Adversarial Client Heterogeneity**" [Under Review at Engineering Applications of Artificial Intelligence (EAAI), Elsevier] arXiv
- **C. Mitra**, *et al.* "**Transfer Learning Based Multiclass Brain Tumor Classification Using MRI Data**," in *Proceedings of IEEE QPAIN*, 2025. [Published at IEEE QPAIN 2025] Paper
- M.M.H.Nipu, **C. Mitra**, *et al.* "**Ben-Misandry: Closing the Bidirectional Gap in Bengali Gender Hate Detection through Transformer Late Fusion**," 2026. [Accepted at IEEE COMPAS 2026]
- M. M. H. Nipu, **C. Mitra**, *et al.* "**PulmoLiteNet: A Memory-Efficient Lightweight Convolutional Network for Accurate Lung Cancer Histopathology Classification on Edge Device**," 2026. [Accepted at IEEE BECITHCON 2026]

Professional Experience

Dexian Bangladesh (ByteMethod AI) Oct 2025 – Present
Application Developer (AI/ML) *Dhaka, Bangladesh*

- **Multi-Agent Product Accelerator (Project BMPA):** Architected an autonomous multi-agent orchestration pipeline using the Claude AgentSDK for automated PRD-to-MVP synthesis (50–70 mins at <$10/run); optimized internal GitLab code component retrieval by restructuring index lookup from $O(n^2)$ to $O(\log n)$.
- **AI-Powered RFP Intelligence Platform:** Engineered an end-to-end compliance analysis and proposal generation pipeline over government tender portals (`sam.gov`) utilizing non-vector hierarchical Page Indexing for long-context retrieval, reducing manual review latency from months to hours at $6 per cycle. Project Page
- **Predictive Operational Workflows (Note2Action):** Designed an enterprise NLP task-prediction pipeline to ingest unstructured weekly logs from 600+ account managers, automating multi-label operational task extraction and workflow delegation. Project Page

Walton Hi-Tech Industries Aug 2025 – Oct 2025
AI Engineer *Dhaka, Bangladesh*

- Architected a production Retrieval-Augmented Generation (RAG) backend in Linux environments to power an enterprise assistant, integrating heterogeneous data sources (HRMS, technical product catalogs) for low-latency context retrieval.

Outlier (Scale AI) Mar 2024 – June 2025
AI Contributor (RLHF & Code Evaluation) *Remote*

- Evaluated LLM reasoning traces and curated high-complexity code generation, refactoring, and debugging datasets to optimize Supervised Fine-Tuning (SFT) and Reinforcement Learning from Human Feedback (RLHF) pipelines.

Technical Skills

Languages: Python, C++, C, SQL, TypeScript, JavaScript, Bash

ML & Deep Learning: PyTorch, TensorFlow, Hugging Face Transformers, Scikit-learn, OpenCV, NumPy, Pandas

LLM & Agentic Systems: DSPy, LangGraph, LangChain, LlamaIndex, PyRIT, Claude AgentSDK, Model Context Protocol (MCP)

Systems, Cloud & Databases: FastAPI, Docker, Kubernetes, Git, Linux, Azure (ML, Container Apps, Bicep), Terraform (IaC), Vector DBs (FAISS, Qdrant, Chroma), PostgreSQL, Snowflake

Selected Academic & Open-Source Projects

Inspecta – Multi-Agent Security & Compliance Auditor | *PyRIT, OWASP, Multi-Agent Systems, Python*

- Engineered a multi-agent security auditing framework utilizing Microsoft PyRIT and static call-graph analysis to automate vulnerability discovery and OWASP compliance verification across web and AI applications.

UI Craft – Model Context Protocol (MCP) Server (Repository) | *MCP, Agentic Tooling, LLMs*

- Developed an open-source Model Context Protocol (MCP) server to standardize context exchange between LLM coding agents and frontend component trees, minimizing structural hallucinations and manual UI overhead.

NLP Predictive Modeling for Academic Paper Retractions (Project Page) | *Scikit-learn, TF-IDF, NLP*

- Built an NLP classification pipeline over retracted scholarly metadata and abstracts using TF-IDF feature extraction and Random Forest, achieving **92%** retraction prediction accuracy alongside unsupervised topic clustering.

4-Bit Custom CPU Microarchitecture (Repository) | *Logisim, Custom ISA, Digital Logic*

- Designed a gate-level 4-bit central processing unit featuring a custom Instruction Set Architecture (ISA), core Arithmetic Logic Unit (ALU) operations, branching logic, and an integrated 8×15 SRAM module.

Mentorship & Teaching Experience

Independent Academic Tutor 2020 – 2024

Computer Science & Mathematics

- Mentored undergraduate and high school students in advanced technical subjects, including Algorithms, Object-Oriented Programming (OOP), Computer Networks, and Introductory Machine Learning.
- Developed tailored curriculum and instructional materials to support university-level coursework and engineering entrance exam preparation.

Achievements

Academic Merit Scholarships: Awarded the Dutch Bangla Bank Scholarship for Undergraduates (2019–2023) and the RUET Technical Scholarship based on Admission Merit (2019, 2022).

Competitive Programming & Algorithmic Problem Solving: Solved 350+ complex algorithmic challenges (200+ on LeetCode, 150+ on Codeforces), demonstrating advanced proficiency in data structure optimization and dynamic programming. Secured 10th position in the RUET Codesmash Intra-University Programming Contest (2019).

University Leadership: Elected Class Representative for four consecutive years, bridging communication between faculty and the student body.

Community Organization: Served as Joint Secretary for the Rajshahi City Association at RUET (2023–2024), overseeing administrative tasks and organizing university events. Worked as volunteer for CSE FEST 2023, Departmental Programs.

References

Dr. Bushra Chowdhury
AI Practitioner Lead *(Current Supervisor)*
Dexian (ByteMethod.ai)
Ph.D. in Computer Science, Virginia Tech
Email: bushra.chowdhury@dexian.com

Dr. Tarek Mahmud
Assistant Professor
Dept. of EECS, Texas A&M University–Kingsville
Ph.D. in Computer Science, Texas State University
Email: tarek.mahmud@tamuk.edu

Farjana Parvin
Assistant Professor *(Undergraduate Thesis Supervisor)*
Dept. of CSE, RUET
Email: farjana@cse.ruet.ac.bd